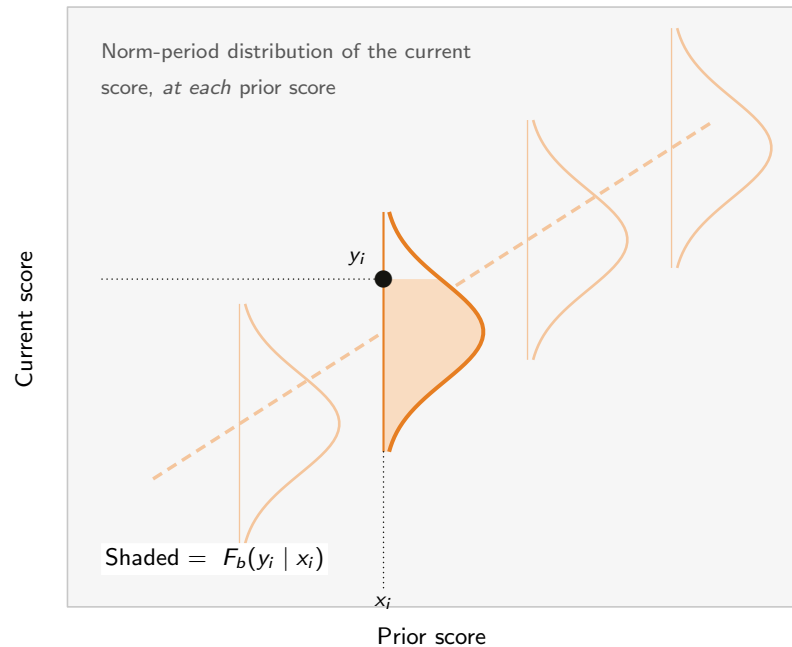


A. Condition on the prior score

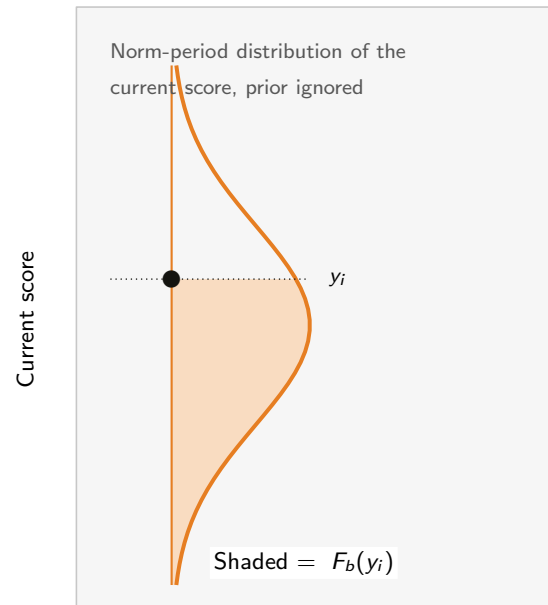
This is the baseline SGP



average over
the prior
 $F_b(y) = E_X[F_b(y | X)]$

B. Drop the conditioning

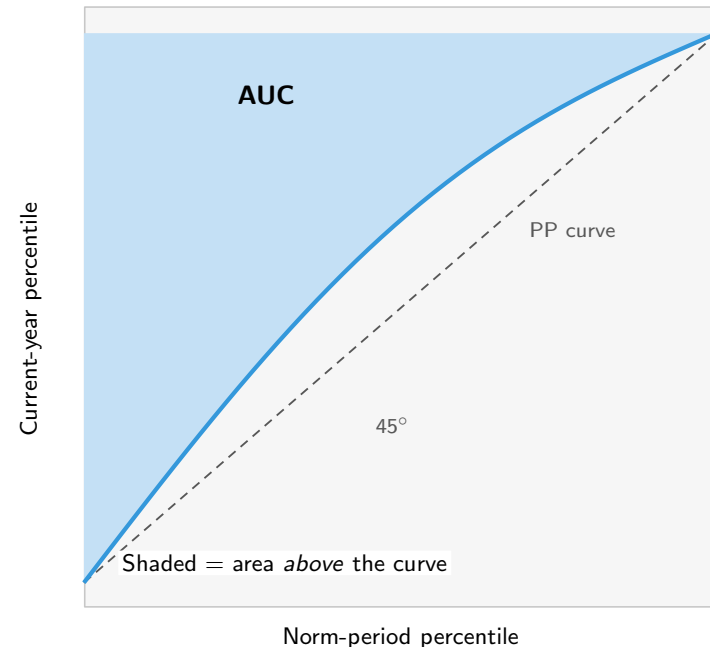
One marginal distribution



average over
the cohort
 $E[F_b(Y_c)]$

C. Average over the cohort

This is Ho's comparison



$$E[F_b(Y_c)] = \Pr(Y_b < Y_c) = \text{AUC}$$

$$V = \sqrt{2} \Phi^{-1}(\text{AUC})$$

A conditions on the prior. B and C do not.